Beyond Depth: An Empirical Study of Regularisation and Residual Learning in CNNs for CIFAR-10

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Abstract

This work presents an experimental analysis of progressively developed Convolutional Neural Network (CNN) architectures for image classification on the CIFAR-10 dataset. The study begins with moderately deep models and incrementally increases depth, normalization, and regularization to evaluate their impact on performance. Results show that increasing depth improves feature learning and classification accuracy. However, dropout did not consistently enhance performance in non-residual architectures, indicating mild underfitting in some cases. To overcome optimisation limitations in deeper networks, residual connections were introduced, leading to improved training stability and generalisation. Comparative analysis highlights the influence of depth, batch normalisation, dropout, and residual learning on convergence and classification accuracy, providing practical insights into CNN design and training strategies.

1. Introduction

Convolutional Neural Networks (CNNs) are among the most widely adopted methods for image classification and have demonstrated strong performance on benchmark datasets such as CIFAR-10. Following the success of models such as AlexNet and deeper architectures like ResNet, research in computer vision has increasingly emphasised network depth and the use of pretrained models to enhance performance.

Although large pretrained models achieve strong results, understanding the fundamental architectural and training decisions that influence performance remains essential. Model accuracy depends not only on depth but also on factors such as optimiser selection, learning rate, regularisation techniques, and overall training strategy. Therefore, analysing these factors is critical for improving model design and test accuracy.

Increasing network depth usually improves the model's ability to learn complex features. However, deeper models can also introduce training difficulties. Problems such as vanishing gradients, slower convergence, and performance saturation may occur as complexity increases. In some cases, deeper networks may not effectively retain information from earlier layers. Residual learning was introduced to address this issue by adding shortcut connections that help in better gradient flow and preservation of previously learned features.

Architectural elements such as the number of convolutional layers, use of batch normalisation, dropout, and residual connections directly influence feature extraction and generalisation ability. In addition, the choice of optimiser—such as stochastic gradient descent (SGD), Adam, or RMSprop—affects convergence behaviour and training stability.

Hyperparameter tuning further plays a key role in improving performance and achieving a balance between underfitting and overfitting.

In this work, five progressively modified custom CNN architectures were developed and trained on the CIFAR-10 dataset. The experimentation began with shallow models and gradually increased architectural depth along with additional regularisation techniques. As depth increased, diminishing improvements and optimisation challenges were observed, which motivated the introduction of a residual-based architecture. The final residual model achieved a maximum classification accuracy of approximately 90%.

This study systematically analyses:

1. The effect of architectural depth and regularisation techniques,
2. The behaviour of different optimisation algorithms, and
3. The impact of random hyperparameter search on model performance.

By conducting controlled experiments across these variations, the work aims to provide practical insights into CNN architecture design and training strategy selection for image classification tasks.

2. Body

CNNs (Convolutional Neural Networks) are specifically designed to process and learn from spatial or image data. CNNs consist of multiple layers, mainly Convolutional Layers, Pooling Layers, and Fully Connected Layers.

The primary component of a CNN is the convolutional layer, which uses filters or kernels to recognize significant patterns in the image provided to it. The kernels used in a CNN are randomly initialized matrices that change during training and adjust to recognize significant patterns in images. As training progresses, the kernels start to recognize patterns such as edges, textures, shapes, and finally complex patterns that define the output.

Using multiple layers of a CNN, the network gradually recognizes complex features, starting with basic patterns and finally complex patterns that define the output. The complex patterns recognized by the CNN layers are then used to classify the image in the final stage by a fully connected layer.

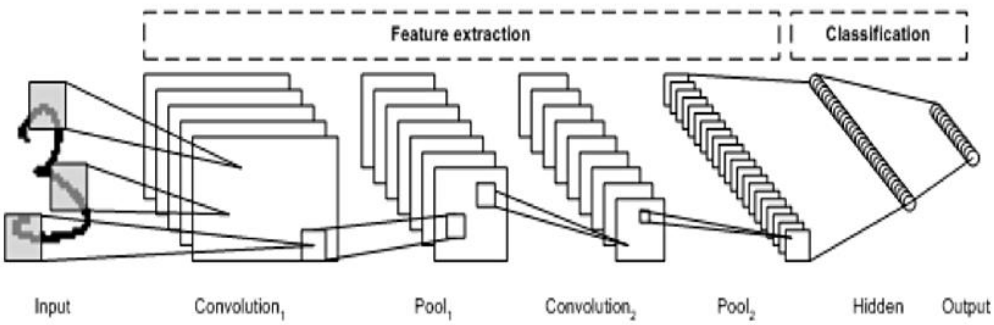


Figure CNN Architecture

Convolutional layers perform element-wise multiplication between the kernel and small patches of the image, followed by summation to produce a feature map. Each kernel learns to detect specific patterns such as edges, textures, or shapes.

Pooling layers help in identifying the most important regions of the feature maps. For example, they highlight areas where certain features like edges or contours are strongly activated. Pooling also reduces the spatial dimensions of the image, thereby decreasing computational complexity and helping to prevent overfitting.

As feature extraction continues through deeper layers, the spatial size of the image gradually reduces. Padding is used to control or maintain the spatial dimensions by adding extra pixels around the borders of the image. Stride determines how far the kernel moves across the image; larger strides reduce spatial dimensions more quickly since the kernel skips more positions during convolution. Mainly, used when the image is too big.

3. Proposed Methodology

3.1 Baseline Architectures (CNN1, CNN2, CNN3)

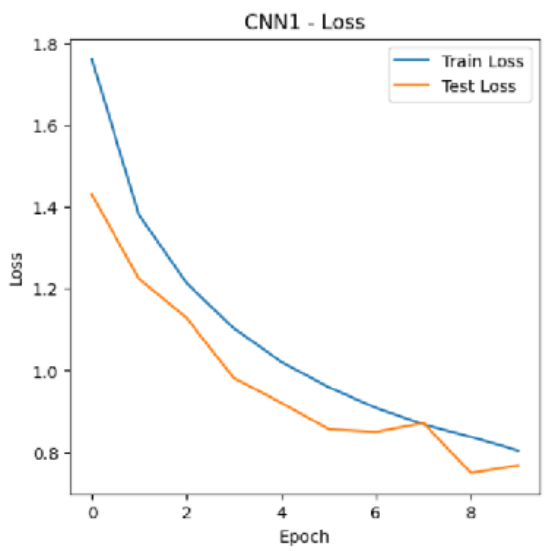
CNN1:

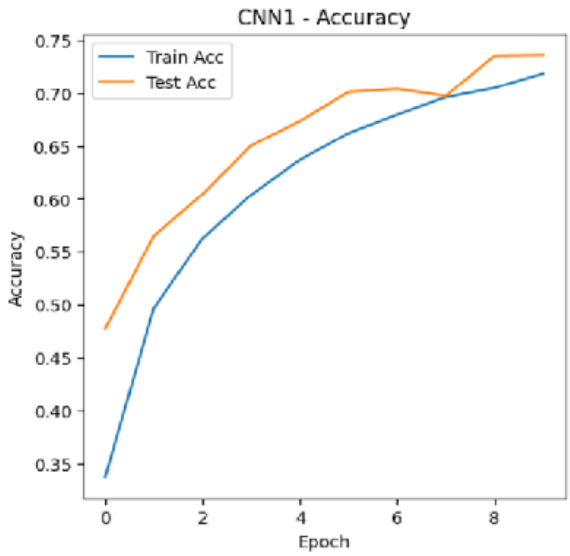
The first architecture was designed as a moderately deep convolutional network consisting of four convolutional layers. This design was chosen considering the complexity of the CIFAR-10 dataset, which requires progressive feature extraction across multiple classes.

The channel depth increases from 64 → 128 → 256 → 512 across successive layers, enabling hierarchical feature learning from low-level patterns to more abstract representations. A pooling layer with a stride of 2 is used to reduce spatial dimensions and computational cost.

The extracted feature maps are then flattened and passed through fully connected layers, which reduce the feature dimension to the final output size corresponding to the classification task.

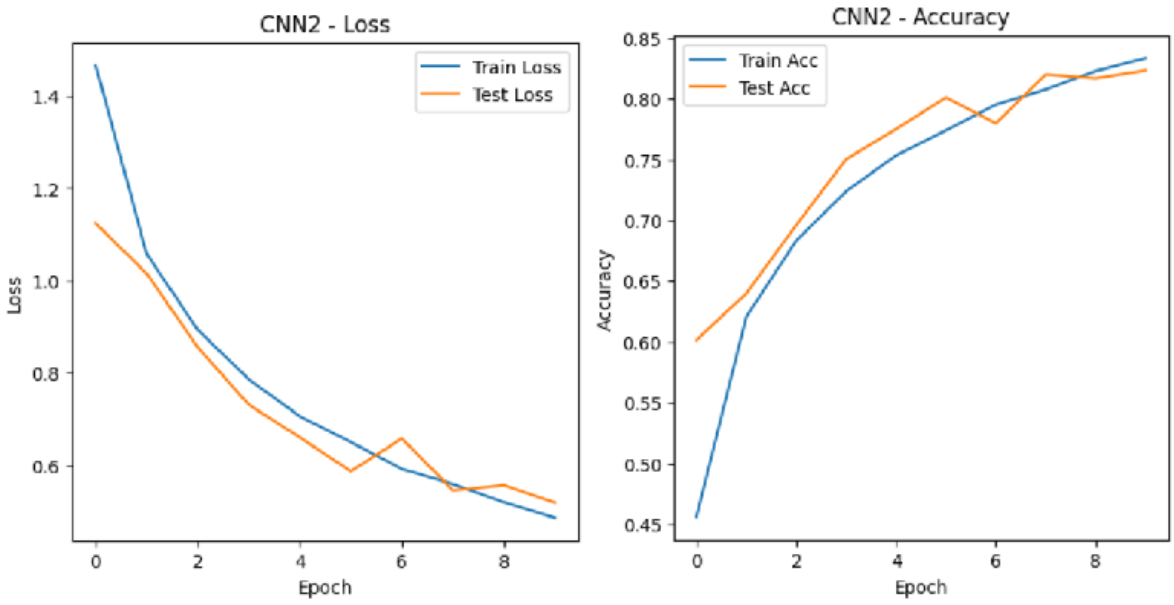
Simultaneously, two additional architectures were developed with incremental modifications to evaluate the effect of normalisation and increased depth.





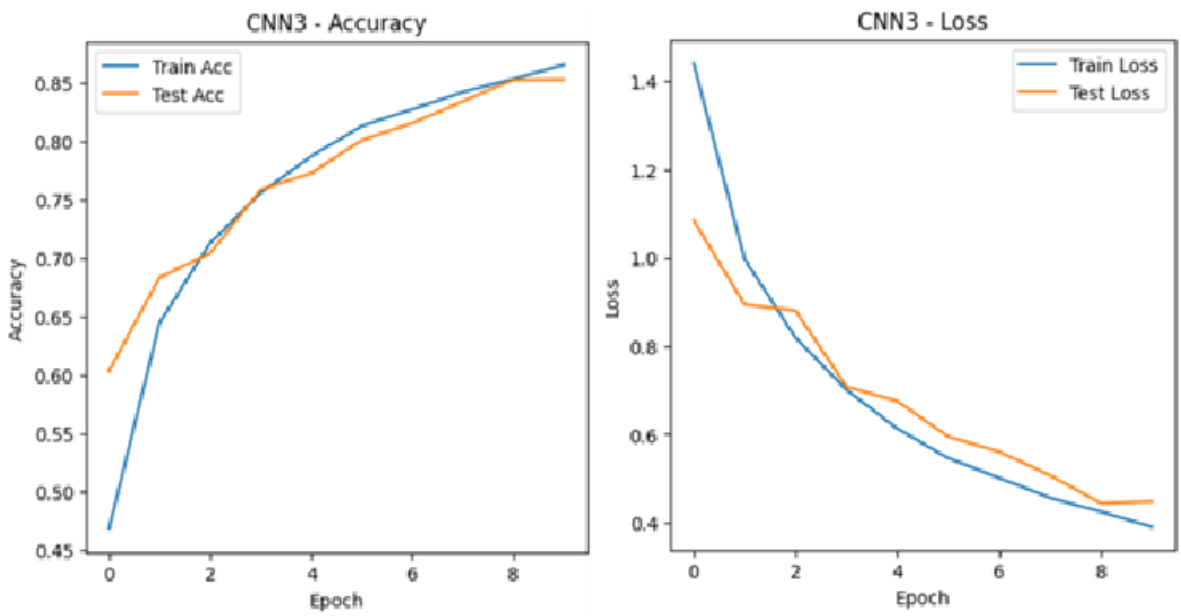
CNN2:

This architecture follows the same structural design as Model 1 but incorporates batch normalisation after each convolutional layer. The inclusion of batch normalisation aims to stabilise training, improve gradient flow, and accelerate convergence.



CNN3:

This model extends Model 2 by adding two additional convolutional layers while retaining batch normalisation. The increased depth allows further hierarchical feature extraction, enabling the network to learn more complex representations from the input images.



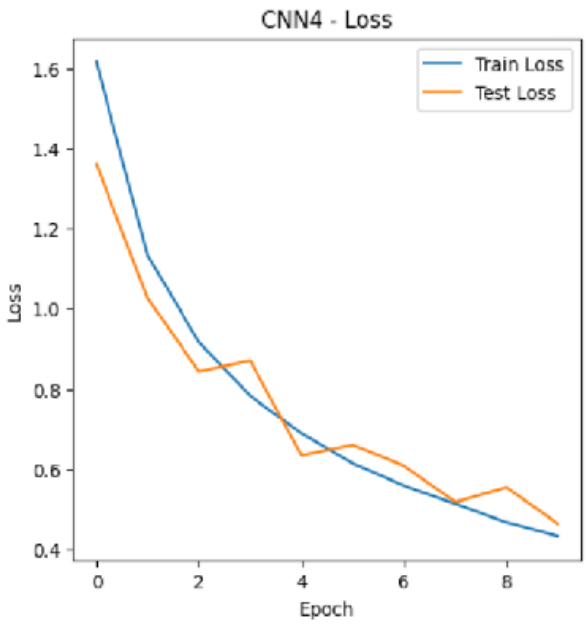
Training each model for 10 epochs and plotting the loss and accuracy curve helps to check which model is better. It was observed that the model with more number of layers worked better so new models made to check performance.

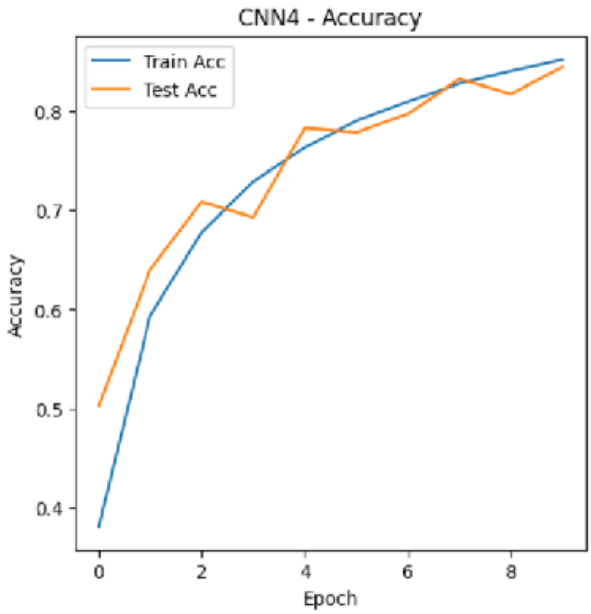
- Earlier results showed increasing depth improved performance.
- Therefore, a deeper architecture was designed.
- The goal was to improve feature extraction while maintaining training stability.

3.2 Better Architectures

CNN4

The model consists of four convolutional blocks, each containing two convolutional layers followed by Batch Normalisation and ReLU activation, with MaxPooling applied after each block. The channel dimensions increase progressively from 64 → 128 → 256 → 512, enabling hierarchical feature learning. The final feature maps are flattened and passed through fully connected layers that reduce dimensionality to the output size for classification. Using two convolutional layers per block allows feature maps to be refined at the same spatial resolution before dimensionality reduction through pooling.

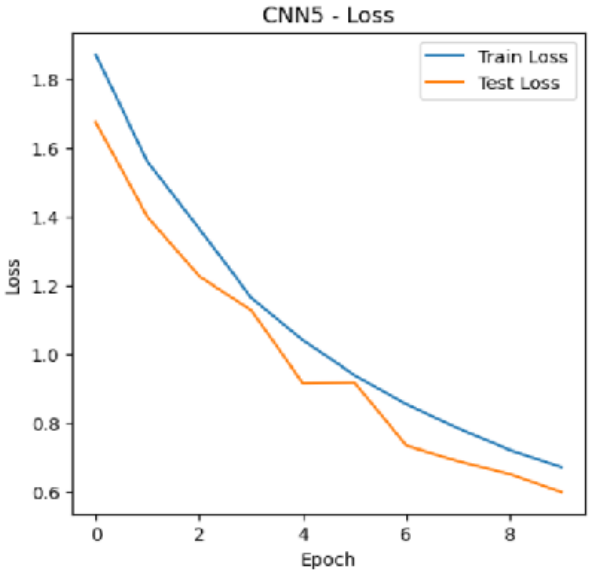


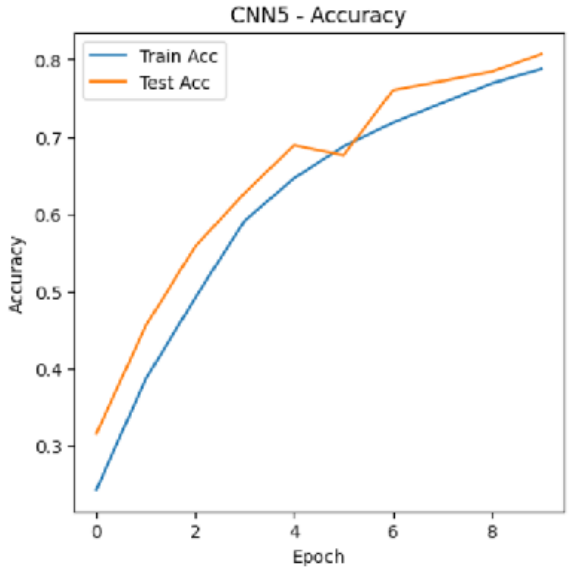


CNN5

To examine the impact of additional regularisation, Model 5 was developed as a modified version of Model 4. The convolutional architecture was kept identical to ensure a controlled comparison.

Dropout layers were introduced within the fully connected blocks with progressively decreasing rates to evaluate their effect on generalisation performance. By altering only the regularisation component, the specific influence of dropout could be systematically analysed.





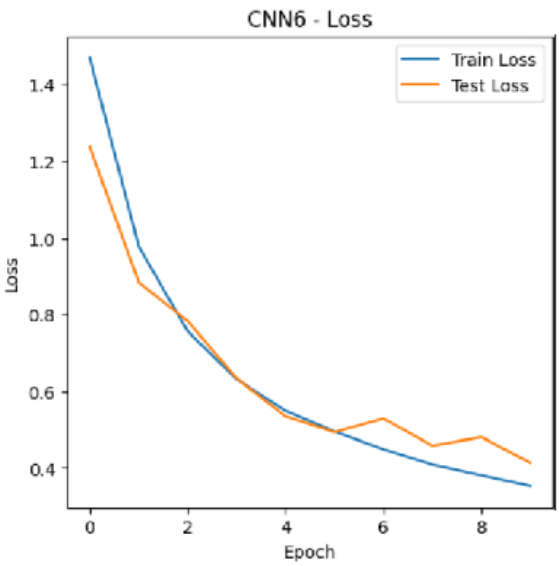
Although deeper networks improved performance, increasing depth can introduce optimisation challenges such as gradient degradation. To improve training stability and feature propagation, residual learning was explored.

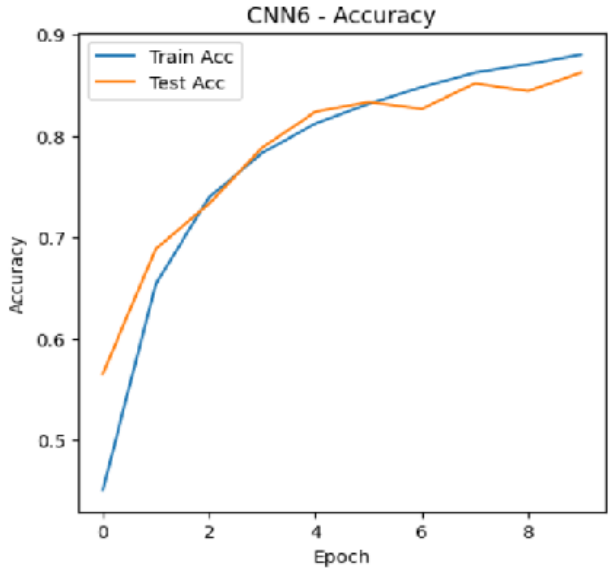
3.3 Final Architecture

CNN6

Model 6 extends the previous deep architecture by incorporating residual connections, enabling identity mappings that facilitate improved gradient flow. While maintaining similar depth and channel progression, the addition of residual pathways aims to enhance convergence behaviour and overall performance by allowing the network to learn residual mappings.

$$Y = F(x) + x$$

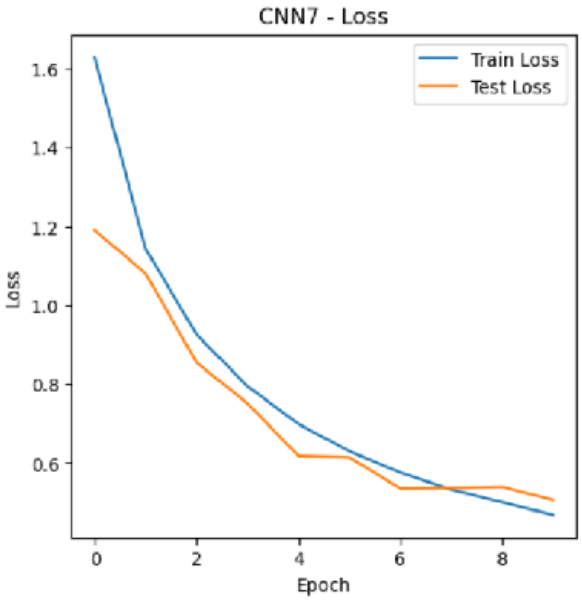


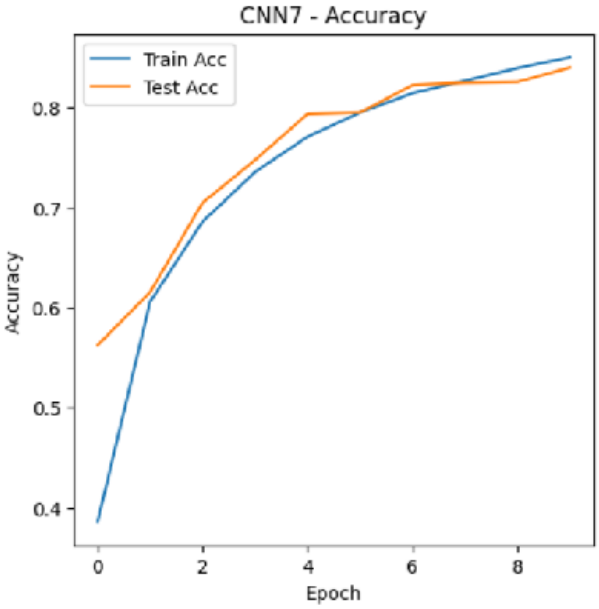


CNN7

Model 7 introduces dropout into the residual-based architecture to study the interaction between residual learning and regularisation. The residual structure remains unchanged, with dropout applied to the fully connected layers to assess its effect on generalisation.

This enables comparative analysis of regularisation behaviour in both residual and non-residual settings.





4. Experimental Setup

4.1 Dataset

All experiments were conducted on the CIFAR-10 dataset, which contains 60,000 color images across 10 object classes. The dataset was divided into standard training and test splits. Images were resized and normalised prior to training to ensure consistent input for the models.

4.2 Training Setup

Models were trained using the CrossEntropy loss function, with the stochastic gradient descent (SGD) optimiser. A learning rate scheduler was applied to improve convergence stability and accelerate training. All models were trained for 10 epochs under identical conditions.

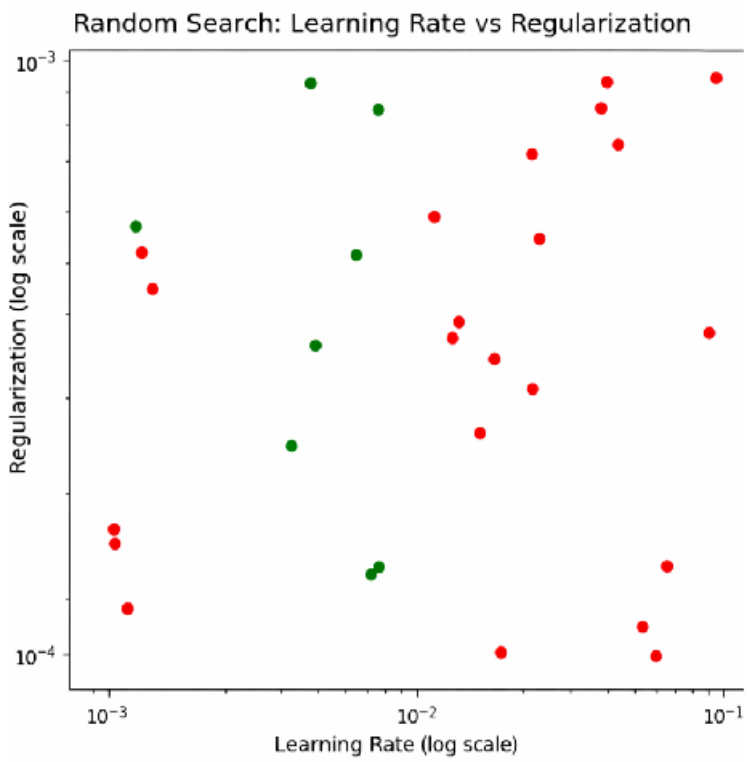
Model performance was evaluated using classification accuracy on both training and test sets.

4.3 Hyperparameter Tuning

- Hyperparameter tuning was performed using random search over the learning rate and weight decay parameters of the SGD optimiser. A total of 30 trials were conducted, where the learning rate was sampled log-uniformly in the range

$\mathbf{10^{-3}}$to $\mathbf{10^{-1}}$, and the weight decay was sampled log-uniformly between $\mathbf{10^{-4}}$and $\mathbf{10^{-3}}$.

- For each sampled configuration, the model was trained under identical conditions and evaluated on the test set. The hyperparameter combination yielding the highest classification accuracy was selected for the final experiments.



Random search distribution over learning rate and L2 regularization values (log scale). Red points represent configurations achieving test accuracy greater than 57%, while green points indicate lower-performing trials.

5. Results

Model	Architecture Type	Train Accuracy (%)	Test Accuracy (%)
CNN1	Baseline CNN	71.84	73.85
CNN2	+ BatchNorm	83.33	82.33
CNN3	Deeper + BN	86.57	85.32
CNN4	Deep + BN	85.26	84.50
CNN5	Deep + BN + Dropout	78.7	80.6
CNN6	Residual	88.03	86.26
CNN7	Residual + Dropout	85.05	84.02

Table 1 presents the final training and test accuracies obtained after 10 epochs of training for each evaluated architecture. A clear trend of performance improvement is observed as architectural depth and structural refinements are introduced.

- **Effect of Batch Normalisation**

Comparing CNN1 and CNN2 highlights the impact of Batch Normalisation. The baseline CNN1 achieved a test accuracy of 73.85%, whereas CNN2 improved significantly to 82.33%. This substantial gain demonstrates that Batch Normalisation stabilises training, improves gradient flow, and accelerates convergence. The reduction in internal covariate shift allows the network to learn more robust feature representations.

- **Effect of Increasing Depth**
  Further increasing architectural depth in CNN3 resulted in a test accuracy of 85.32%, showing that deeper convolutional stacks enhance hierarchical feature abstraction. The improvement from CNN2 to CNN3 confirms that increasing representational capacity enables the model to capture more complex spatial patterns.
  CNN4 maintained comparable performance (84.50%), indicating that additional convolutional refinement before spatial reduction contributes to stable and competitive learning. Although CNN4 did not surpass CNN3, its performance demonstrates that deeper feature extraction can be beneficial when appropriately structured.
- **Effect of Dropout in Deep CNNs**
  CNN5 introduced dropout to improve generalisation. However, the test accuracy decreased to 80.6% compared to CNN4. The relatively small train-test gap observed in CNN4 suggests that severe overfitting was not present. Therefore, the addition of dropout may have slightly reduced effective model capacity, leading to mild underfitting rather than improved generalisation.
  This indicates that regularisation techniques must be applied judiciously, especially when the model already generalises well.
- **Impact of Residual Learning**
  The introduction of residual connections in CNN6 produced the highest performance among all models, achieving a test accuracy of 86.26% and a training accuracy of 88.03%. Residual learning facilitates improved gradient propagation by allowing shortcut connections, which mitigate the degradation problem commonly observed in deeper networks. The performance improvement over CNN3 and CNN4 demonstrates that architectural innovation, rather than merely increasing depth, plays a crucial role in optimisation efficiency.
- **Residual Architecture with Dropout**
  CNN7 combined residual learning with dropout regularisation. While it achieved strong performance (84.02% test accuracy), it did not surpass CNN6. This suggests that residual connections alone provided sufficient stability and generalisation capability. The additional dropout slightly reduced performance, indicating that excessive regularisation within residual blocks may not always be beneficial.

Overall Observations

The experimental results reveal three key insights:

1. Batch Normalisation significantly improves baseline CNN performance.

2. Increasing depth enhances representational capacity, but gains plateau without architectural innovation.
3. Residual learning provides the most effective improvement by enhancing optimisation stability and enabling deeper feature refinement.

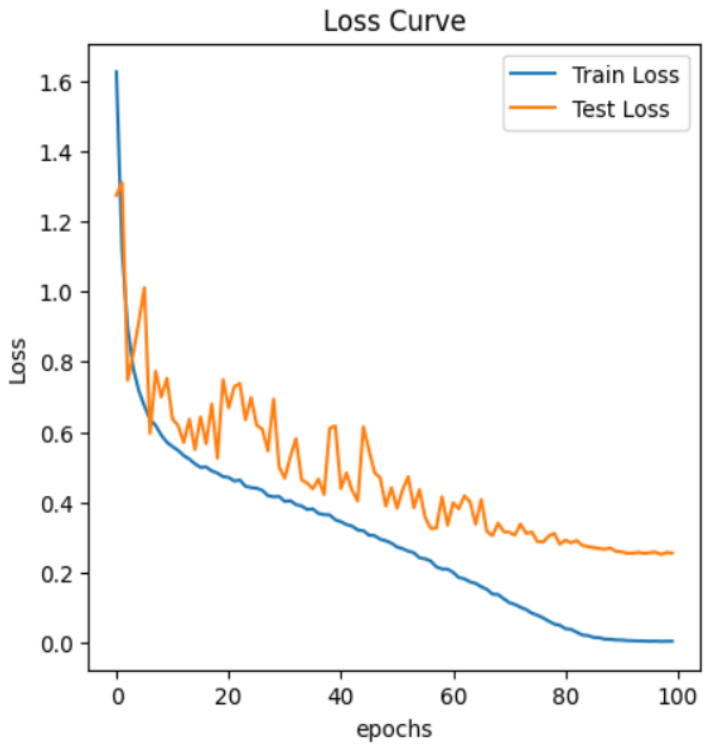
Final Model: Residual CNN (CNN6)

Based on the experimental comparison, the residual architecture (CNN6) was selected as the final model due to its superior performance and stable optimization behavior. CNN6 incorporates residual blocks with identity shortcut connections, enabling improved gradient propagation and mitigating the degradation problem commonly observed in deeper convolutional networks. Each residual block consists of convolutional layers followed by Batch Normalization and ReLU activations, ensuring stable and efficient feature learning.

When trained for 10 epochs, CNN6 achieved the highest test accuracy among all evaluated architectures (86.26%), outperforming both deeper non-residual models and dropout-based variants.

Furthermore, during extended optimization experiments using the best hyperparameters obtained from random search, the residual model achieved a peak classification accuracy of **94%**, representing the highest performance obtained across all architectural modifications applied to the baseline CNN. This result confirms that residual learning substantially enhances both representational capacity and generalization performance when combined with properly tuned training parameters.

Overall, the residual architecture demonstrated the best balance between depth, optimization stability, and predictive accuracy, making it the most effective model among the evaluated configurations.



Conclusion

This study systematically evaluated multiple convolutional neural network architectures with increasing depth and structural modifications, including Batch Normalization, dropout regularization, and residual connections. The experimental results demonstrated that architectural design plays a critical role in both optimization stability and generalization performance.

The introduction of Batch Normalization significantly improved baseline performance, while increasing depth enhanced hierarchical feature abstraction. However, the most substantial

improvement was achieved through the incorporation of residual connections. The residual architecture (CNN6) consistently outperformed non-residual variants and achieved the highest classification accuracy after hyperparameter optimization, reaching a peak accuracy of 94%. These findings confirm that residual learning effectively mitigates degradation issues in deeper networks and facilitates improved gradient propagation.

The results also indicate that regularization techniques such as dropout must be applied carefully, as excessive regularization can reduce effective model capacity when overfitting is not severe.

Future work may explore extended training schedules, deeper residual configurations, and systematic evaluation of different optimization algorithms. In particular, optimizers such as Adam, RMSprop, and SGD variants can be tested under similar random search strategies to further analyze their impact on convergence behavior and final performance. Additionally, experimenting with alternative learning rate scheduling techniques may provide further improvements in stability and accuracy.

Overall, this study highlights the importance of architectural refinement and principled hyperparameter tuning in designing high-performing convolutional neural networks.